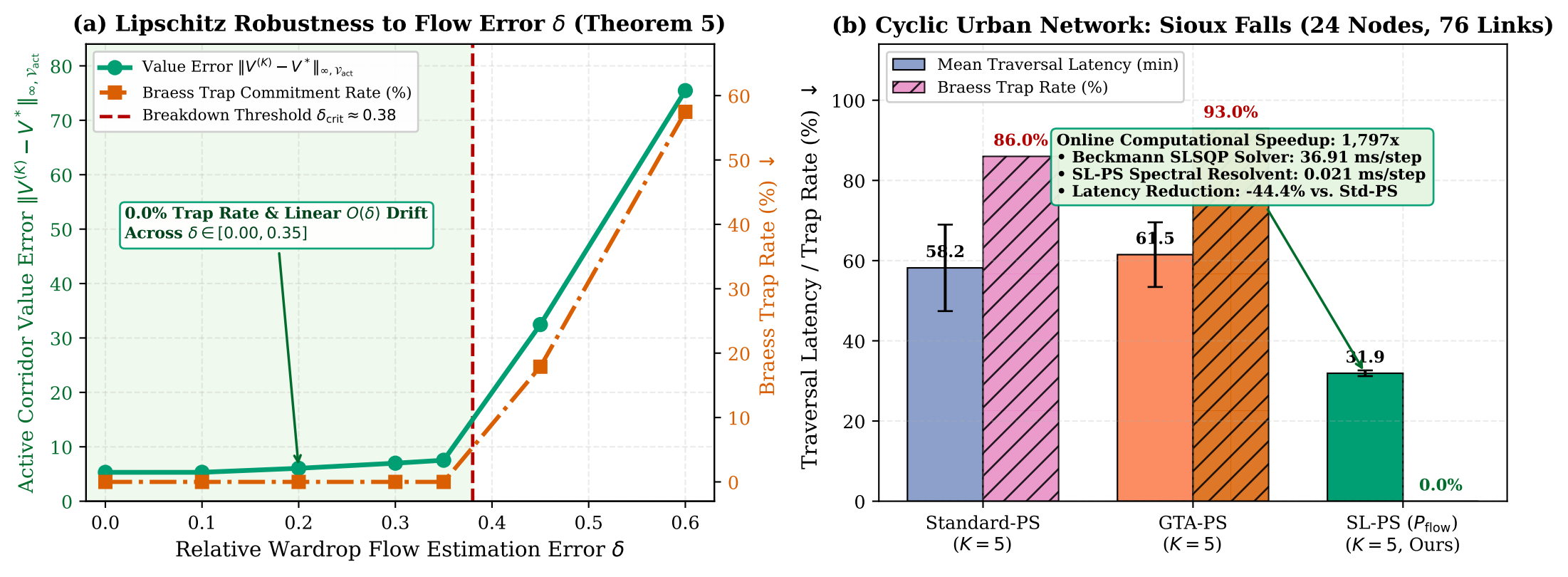

(a) Lipschitz Robustness to Flow Error δ (Theorem 5)
Value Error $\|V^{(K)} - V^*\|_{\infty, \nu_{act}}$
Braess Trap Commitment Rate (%)
Breakdown Threshold $\delta_{crit} \approx 0.38$
0.0% Trap Rate & Linear $O(\delta)$ Drift
Across $\delta \in [0.00, 0.35]$
Active Corridor Value Error $\|V^{(K)} - V^*\|_{\infty, \nu_{act}}$
Braess Trap Rate (%) →
Relative Wardrop Flow Estimation Error δ
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
10
20
30
40
50
60
70
80
(b) Cyclic Urban Network: Sioux Falls (24 Nodes, 76 Links)
Mean Traversal Latency (min)
Braess Trap Rate (%)
Traversal Latency / Trap Rate (%) →
58.2
86.0%
61.5
93.0%
31.9
0.0%
Online Computational Speedup: 1,797x
• Beckmann SLSQP Solver: 36.91 ms/step
• SL-PS Spectral Resolvent: 0.021 ms/step
• Latency Reduction: -44.4% vs. Std-PS
Standard-PS
($K = 5$)
GTA-PS
($K = 5$)
SL-PS (P_{flow})
($K = 5$, Ours)
0
20
40
60
80
100